Lecture 10

GPS, Higher Dimensions and Waves

Instructor: Geo Jose

GPS

Before we conclude our discussion on Special Relativity, it is worth emphasizing that the equations we have derived over the past few weeks are far from being merely abstract textbook exercises. These things affect us. Consider for example, the case of the case of United States NAVSTAR Global Positioning System (GPS for short). The standard GPS system is designed with 24 satellites in high Earth orbit, though as many as 30 may be operational at a time.

Each satellite orbits at an altitude of about 20,000 km above the Earth's surface, moving at roughly 14,000 km/h. Each satellite carries with it an atomic clock that tickswith a nominal accuracy of 1 nanosecond. A GPS receiver establishes its position and trajectory by analyzing the timing signals it picks up from the visible GPS satellites typically 6 to 12 at a time. The accuracy is remarkable: even a basic handheld GPS device can determine one's position on Earth's surface to within 5–10 meters in just a few seconds. To achieve this level of precision, the clock ticks from the GPS satellites must be known to an accuracy of 20-30 nanoseconds. Because an observer on the ground sees the satellites in motion relative to them, Special Relativity predicts that we should see their clocks ticking more slowly. Special Relativity predicts that the on-board atomic clocks on the satellites should fall behind clocks on the ground by about 7 microseconds per day.

There is an even more inportant issue: because GPS satellites orbit high above Earth, they experience less spacetime curvature than clocks on the surface. General Relativity predicts that clocks nearer to a massive object tick more slowly, so from Earth's perspective, satellite clocks run slightly faster. The expected difference is about 45 microseconds per day. When both relativistic effects are combined, GPS satellite clocks end up ticking about 38 microseconds per day faster than clocks on Earth (since 45 – 7 = 38). While this might sound tiny, GPS demands nanosecond precision—and 38 microseconds equals 38,000 nanoseconds. If left uncorrected, the system's position estimates would already be wrong after only two minutes, and the error would grow by roughly 10 kilometers per day. Without these corrections, GPS would quickly fail as a navigation tool.

Higher Dimensions:

So far we have been considering a one-dimensional world: Observer O (who measures coordinates (x, t)) at rest and observer O$'$ (who measures coordinates (x', t')) moving along the positive x-axis with speed u. We found that these coordinates are realted by Lorentz transformations:

$$x' = \gamma(x - ut) \tag{1}$$

$$t' = \gamma\left(t - \frac{ux}{c^2}\right). \tag{2}$$

If we now include the y and z axis, these coorinates are unaffected by motion along x direction:

$$y' = y \tag{3}$$

$$z' = z. \tag{4}$$

The case of velocities is a bit more interesting. Above equations can written in the following form (take two space-time events: (x_1, y_1, z_1, t_1) and (x_2, y_2, z_2, t_2)) and take their difference):

$$\Delta x' = \gamma(\Delta x - u\Delta t) \tag{5}$$

$$\Delta t' = \gamma\left(\Delta t - \frac{u\Delta x}{c^2}\right) \tag{6}$$

$$\Delta y' = \Delta y \tag{7}$$

$$\Delta z' = \Delta z. \tag{8}$$

Components of velocities measured by O′ is $\frac{\Delta x'}{\Delta t'} = v'_x$ and O is $\frac{\Delta x}{\Delta t} = v_x$ and so on. Taking ratios of the equations above, we see that

$$v'_x = \frac{v_x - u}{1 - \frac{uv_x}{c^2}} \tag{9}$$

$$v'_y = \frac{v_y}{\gamma\left(1 - \frac{uv_x}{c^2}\right)} \tag{10}$$

$$v'_z = \frac{v_z}{\gamma\left(1 - \frac{uv_x}{c^2}\right)}. \tag{11}$$

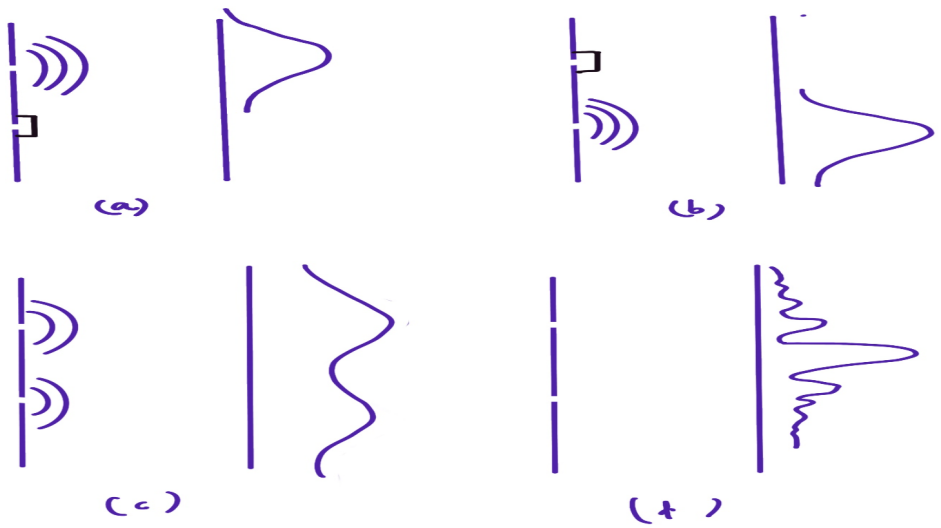


Figure 1:

Waves:

We are going to describe some phenomena that are characteristic of waves. Many different types of waves are familiar to us from everyday life. Waves in water, waves on a string, sound waves, light etc. All of these waves obey the wave equation:

$$\frac{\partial^2\psi(x,y,z,t)}{\partial x^2}+\frac{\partial^2\psi(x,y,z,t)}{\partial y^2}+\frac{\partial^2\psi(x,y,z,t)}{\partial y^2}-\frac{1}{v^2}\frac{\partial^2\psi(x,y,z,t)}{\partial t^2}=0. \tag{1}$$

Here $\psi(x)$ could represent the height of various parts of a string, could be the water level at various positions in space measured from a reference level, it could be the deviations in the air pressure, it could be electric or magnetic fields etc., the contexts could be different but all of those waves obey the wave equation. This equation is linear. If you have two solutions, a linear combination of those will also be a solution.

We shall first discuss the phenomenon of interference. You all have some idea of interference from class 12. Waves from two sources combine to produce interesting patterns. Here is a set up: There is a wall and light is coming in from the left side of the wall. On this wall is a tiny circular hole, and light passes through this hole. Now, on the right side of this wall is another wall on which the light passing though the hole falls upon. What will you see on second wall? You will see a circular shaped region on the wall. However if you reduce the radius of the hole on the wall, you will see that the the circular region on the wall increases in radius; light fans out. That itself is not in agreement with ray optics that you have used to find the positions of images froma mirror, lens etc.

If you make another hole, there are now two tiny holes in the wall through which a tiny bit of light goes through. Close the first hole, you'll see what we saw above in front of the second hole (see Fig 1 (a)).

Now close the second hole you see light in front of the first hole (see Fig 1 (b)). But open both: we expect to see light in front of both holes and the screen brightness to be a sum of what we saw in the above cases as shown in Fig 1(c). And that is what happens usually: if we open a window to get some sunlight, and then open one more, we get more sunlight coming through both the holes. We feel warmer with two windows open than with one open.

Under certain conditions, this does not happen; what we get it is a pattern as shown in Fig 1 (d). That is due to the phenomenon of interference. It is quite mysterious: there are some points on the screen where you get less light, if you open one more slit! In other words there are points on the screen at which there was light with one slit open, but no light at all when one more slit is opened. The conditions are: (1) The wavelength λ of the light should be comparable to the radius of the slits and the distance between the slits. (2) If you make the slits and distances very small and do this with white light, you still will not see these maxima and minima. You need light of a single wavelength to see this.

Next time, we will see why this is the case!